Futures Time-Series Momentum

ORIE 5260 · Midterm Project Presentation

Data validation, strategy construction, attribution, and risk forecasting

58 futures markets · Data: 1969–2014

Main evaluation: 1976–2014

Based on midterm_report.md, project code, and existing outputs

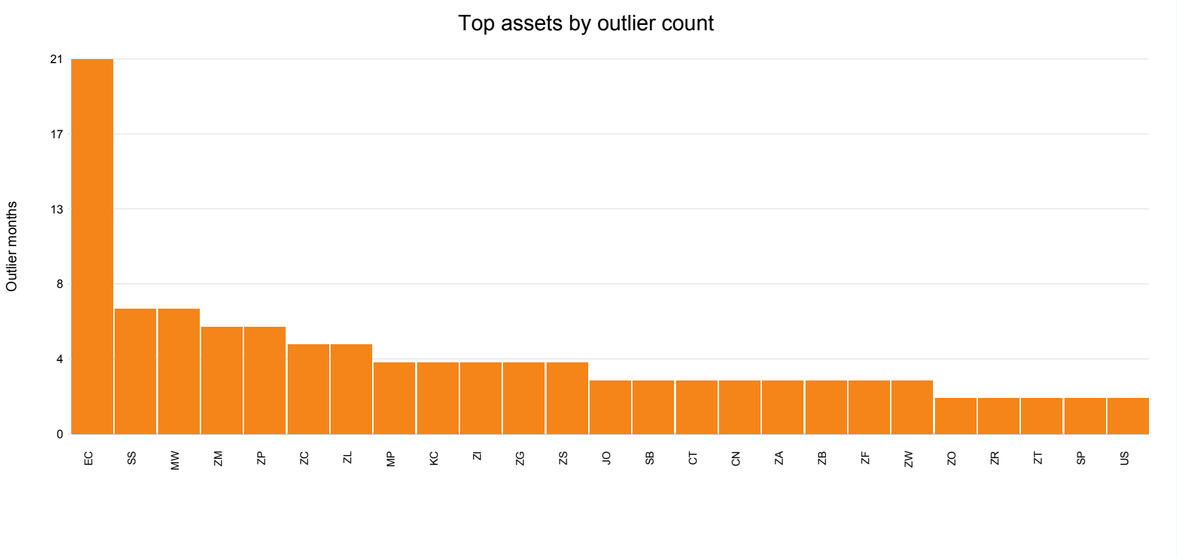
Research question and data

Do past return directions predict next-month futures returns?

Universe	Data
58 markets	26 commodities, 14 equities, 10 fixed income, 8 FX
1969–2014	552 monthly dates; 20,987 valid observations
Unequal histories	551 months at the longest; 75 at the shortest

We evaluate two baselines, then examine returns, risk, and improvements.

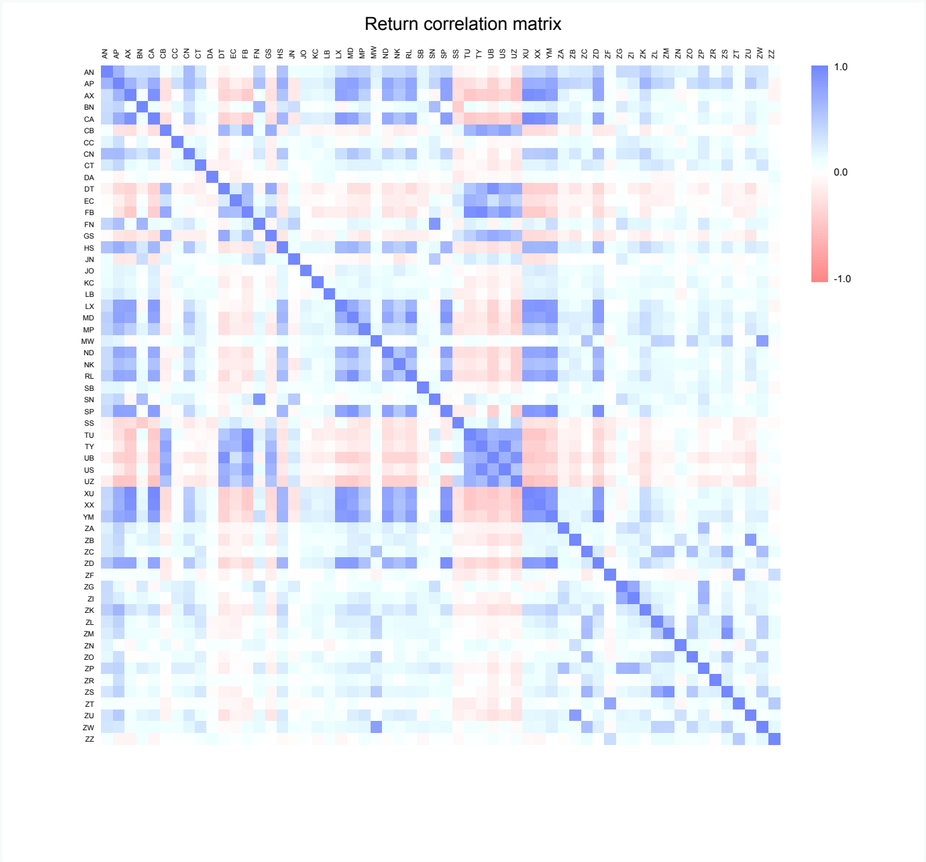
Data quality: flags, reconstruction, and open checks



- 126 outliers flagged with 3×IQR or |robust z| > 5; retained for review.
- 14,974 daily-to-monthly overlapping checks; typical differences approximately zero.
- RL maximum discrepancy: 4.70 percentage points, unresolved.
- No history backfill; missingness includes different start and end dates.

analysis_eda.py; outliers.csv; recomputation_check.csv

EDA implications for portfolio design



- Median annual vol: commodities 29.1%, fixed income 4.8%.
- Within-class correlation: equities 0.76, fixed income 0.62.
- Unequal histories require dynamic eligibility.
- Volatility scaling balances standalone risk; covariance still matters.

EDA outputs; additional summaries of MonthlyReturns.csv

Two baseline versions: rolling and course PPT

Dimension	Rolling baseline	PPT / adjusted baseline
Script	rolling_backtest.py	strategy_backtest_adj.py
12-month signal	sign[∏(1+r) − 1]	sign[Σr]
Volatility estimate	Trailing 12 months	Expanding; at least 36 valid months
Evaluation sample	Feb 1970–Dec 2014; 539 months	Jan 1976–Dec 2014; 468 months
TSMOM CAGR / Sharpe	19.12% / 1.06	10.64% / 0.97
TSMOM vol / MaxDD	18.05% / −27.58%	11.15% / −26.03%

Both use historical inputs, execute in the next month, and compound wealth. Different signals, risk windows, eligibility, and samples prevent a signal-only comparison.

rolling_backtest_output/performance_summary.csv: rolling_tsmom, 0 bps; strategy_backtest_adj_output/baseline_summary.csv

The course PPT approach: adjusted baseline

$$s_{i,t} = \operatorname{sign}\left(\sum_{k=011} r_{i,t-k}\right)$$

$$\hat{\sigma}_{i,t} = \operatorname{Std}(r_{i,1:t}) \times \sqrt{12}$$

$$w_{i,t} = \frac{1}{N_t} \times s_{i,t} \times \frac{0.40}{\hat{\sigma}_{i,t}}$$

- Expanding volatility requires at least 36 valid monthly observations.
- Weights formed in month t earn returns in t+1; rebalance monthly.
- Static uses the same volatility and eligibility, always long.

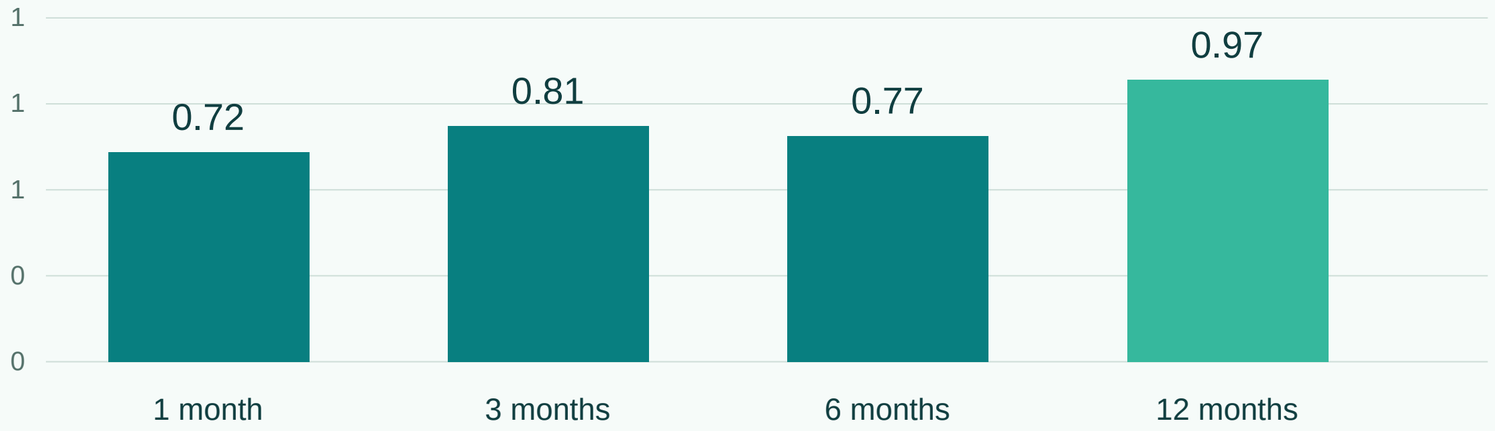
strategy_backtest_adj.py follows the course PPT approach. The 40% parameter scales individual markets; it is not a portfolio volatility target.

Execution timing and assumptions

Formation month t	Execution month t+1
Signal and risk estimates use data through t	execution_weights = weights.shift(1)
Construct equally allocated inverse-volatility positions	Rp,t□1 = Σ wi,t ri,t□1
Keep formation and execution alignment separately	Timing checks confirm a one-month lag

- Missing realized returns for held positions are filled with zero.
- No leverage cap, margin constraints, financing, or explicit roll costs.
- The diagonal risk model assumes zero cross-market covariance.

Four lookbacks: complete portfolio backtests



Lookback	1 month	3 months	6 months	12 months
CAGR	7.71%	9.23%	8.47%	10.64%
Annual vol	11.23%	11.81%	11.36%	11.15%

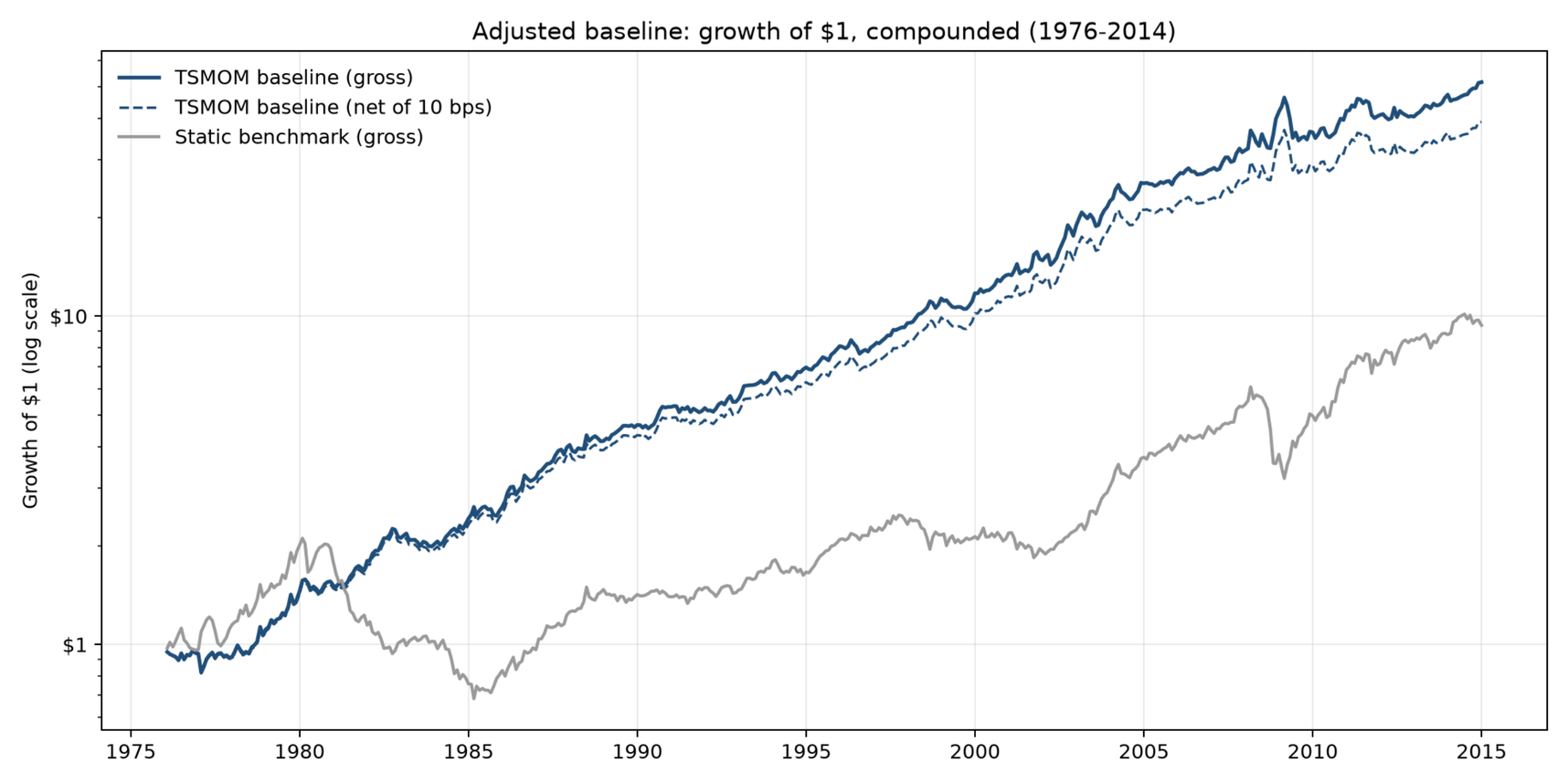
strategy_backtest_adj_output/lookback_summary.csv; 1976–2014; RF = 0

Main results: stronger returns and smaller drawdowns

1976–2014; RF = 0	PPT / adjusted TSMOM	Static long
CAGR	10.64%	5.91%
Annualized mean	10.77%	6.82%
Annualized volatility	11.15%	14.52%
Sharpe: annualized mean / vol	0.97	0.47
Maximum drawdown	−26.03%	−67.54%
Positive-return months	64.1%	60.0%
Annual turnover	7.24	0.41

baseline_performance_summary.csv□strategy_backtest_adj_output/baseline_summary.csv

Compounded wealth of the main strategy



outputs/baseline_performance_output/baseline_cumulative_return.png

Adjusted baseline: exposure and trading costs

Risk / implementation	Evidence
Gross exposure	Average 3.13; maximum 4.60
Net exposure	Average +0.84
Annual turnover	7.24 versus static 0.41
0 / 10 / 25 bps CAGR	10.64% / 9.85% / 8.68%
0 / 10 / 25 bps Sharpe	0.97 / 0.90 / 0.81

Costs are illustrative; leverage, financing, roll, and liquidity constraints remain unmodeled.

Adjusted baseline versus static: correlation

1976–2014, gross	Result
Overall TSMOM–static correlation	0.080
Class correlations: COMM / EQ / FI / FX	0.211 / −0.119 / 0.684 / −0.123
Annual mean TSMOM minus static	3.95 percentage points

Low correlation means different return behavior. It does not alone establish outperformance or crisis protection.

Rolling baseline: a second strategy specification

Component	Implementation in rolling_backtest.py
Signal	Sign of compounded trailing 12-month return
Volatility	Trailing 12 valid consecutive months × √12
Eligibility	Valid signal, valid positive volatility; matched static universe
Position	(1/N) × signal × 0.40 / volatility
Execution	Formation-month weights shifted one month
Evaluation	Feb 1970–Dec 2014; 539 months

Compounded versus summed momentum signals

Both baselines compound wealth; their historical signal aggregation differs

Example: +20%, then −18%	Calculation	Position direction
Rolling compound signal	1.20 × 0.82 − 1 = −1.6%	Short
Adjusted sum signal	20% − 18% = +2.0%	Long

- Large swings can cause opposite signal directions.
- Risk-window and eligibility differences also affect portfolio results.
- A controlled signal-only test would hold all other choices fixed.

rolling_backtest.py signal usage; strategy_backtest_adj.py: momentum_signal; illustrative example

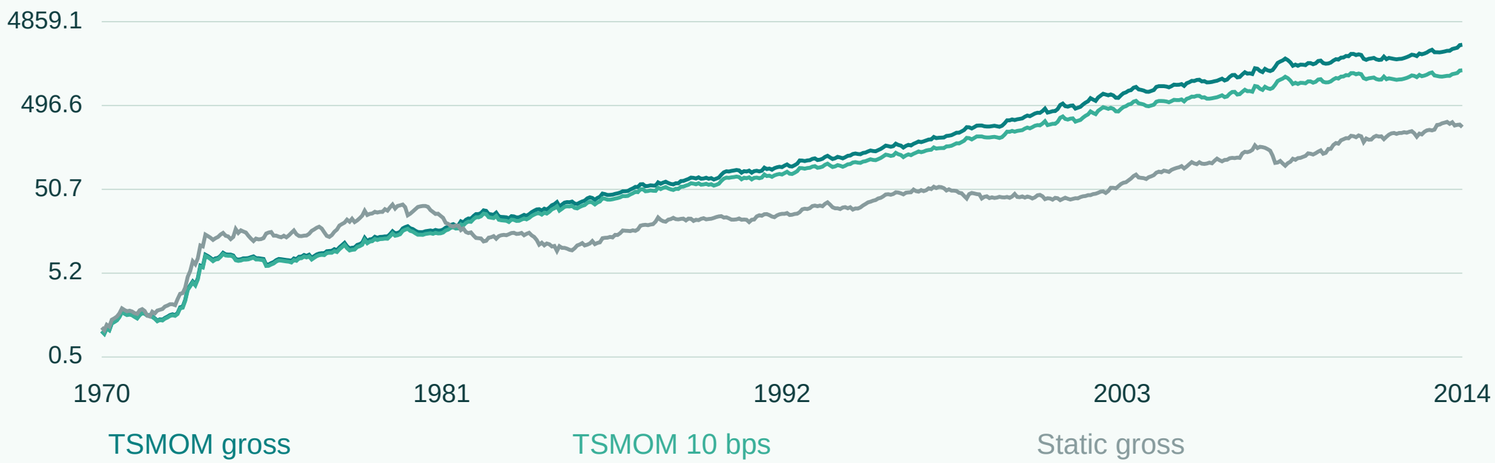
Rolling baseline: performance and cost sensitivity

Rolling baseline, Feb 1970–Dec 2014	CAGR	Annual vol	Sharpe	MaxDD
TSMOM, 0 bps	19.12%	18.05%	1.06	−27.58%
TSMOM, 5 bps	18.21%	18.05%	1.02	−27.76%
TSMOM, 10 bps	17.30%	18.06%	0.97	−27.93%
TSMOM, 25 bps	14.60%	18.12%	0.84	−28.45%
Static, 0 bps	13.33%	21.89%	0.68	−71.78%

- Compounded 12-month signal with trailing 12-month volatility.
- TSMOM and static share the same eligible universe.
- Average monthly traded notional: 1.30; compare costs separately from the PPT baseline.

rolling_backtest.py; rolling_backtest_output/performance_summary.csv: rolling_tsmom / rolling_static rows only

Rolling baseline: compounded wealth



Log wealth scale. These curves use only rolling-baseline return columns.

Rolling correlation with its static benchmark

Cost applied to both portfolios	Correlation	Annual mean TSMOM − static
0 bps	0.3846	4.29pp
5 bps	0.3839	3.77pp
10 bps	0.3831	3.26pp
25 bps	0.3805	1.71pp

- Correlation is moderate and stable across these cost scenarios.
- Positive mean differences remain, but shrink with trading costs.
- This is descriptive return difference, distinct from regression alpha.

rolling_backtest_output/portfolio_correlations.csv

Rolling regressions: benchmark and factor evidence

Regression	Gross	10 bps
Static regression annual alpha	14.43%	13.05%
Static beta / R^2	0.317 / 14.79%	0.316 / 14.68%
Six-factor annual alpha, 2000–2014	9.12%	6.38%
Six-factor alpha p-value	0.0116	0.0862

HAC standard errors use 3 lags. Regression intercepts differ from simple return differences; factor controls are incomplete.

Why do the static correlations differ?

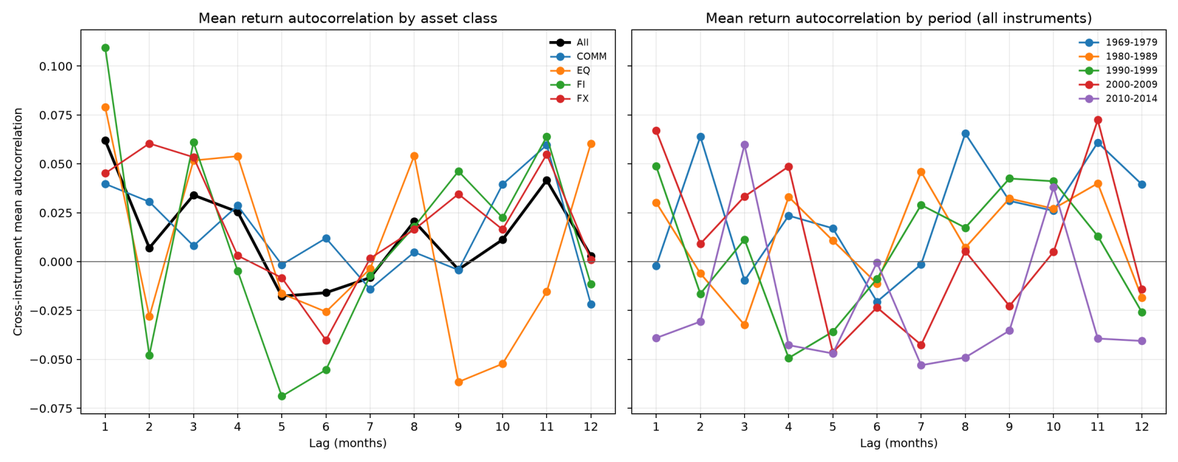
Comparison	PPT / adjusted	Rolling
Corr with own static	0.080	0.385
Signal / volatility	Sum / expanding	Compound / trailing 12m
Sample	1976–2014	1970.02–2014

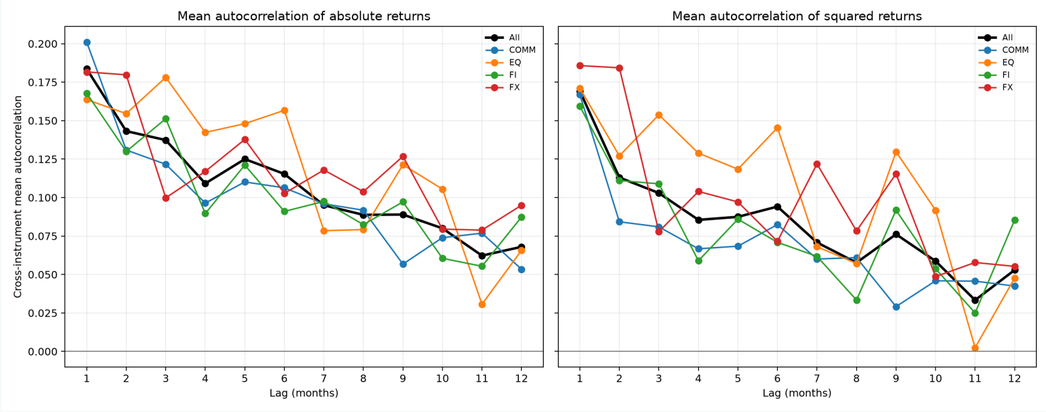
- Possible mechanism: shared recent-volatility scaling may increase co-movement between rolling TSMOM and static.
- Signal signs, eligibility, and sample periods also differ; their effects are not isolated.
- Each static benchmark is different. Lower correlation alone does not imply a better strategy.

Observed gap, untested explanation. Align samples and eligibility, then change one component at a time.

Adjusted baseline_summary.csv; rolling portfolio_correlations.csv; mechanism is a hypothesis

Autocorrelation: returns versus volatility





ACF(k) = Corr(xt, xt□k). Test k = 1–12 for returns, |returns|, and returns². Mean lag-1: raw 6.2%, absolute 18.4%.

Signal predictiveness across four horizons

Lookback	Next r after +	Next r after −	Spread	Hit rate	HAC t*
1m	0.56%	-0.02%	0.58%	52.80%	3.61
3m	0.65%	-0.14%	0.79%	53.46%	4.10
6m	0.60%	-0.12%	0.72%	54.02%	3.73
12m	0.59%	-0.20%	0.79%	54.40%	4.51

Spread = E[rt□1 | st > 0] − E[rt□1 | st < 0]. Hit rate = share of nonzero signals matching next-return direction. *HAC t uses monthly aggregated vol-scaled signed returns.

signal_predictiveness_by_lookback.csv; signal_diagnostics.py

Signal stability by class and period

12m signal	Hit rate	Spread
Commodities	52.8%	0.97%
Equities	56.9%	0.93%
Fixed income	56.8%	0.19%
FX	55.9%	0.44%

Period	Hit rate	Spread
1969–1979	54.0%	1.72%
1980s	54.3%	0.89%
1990s	54.6%	0.77%
2000s	54.7%	0.73%
2010–2014	53.8%	0.11%

Instrument-level diagnostics are aggregated by class and period. Similar hit rates can coexist with smaller economic return spreads.

signal_predictiveness_by_asset_class.csv; signal_predictiveness_by_period.csv; instrument and period ACF outputs

Rolling performance and drawdown recovery



Worst adjusted drawdown: −26.03%

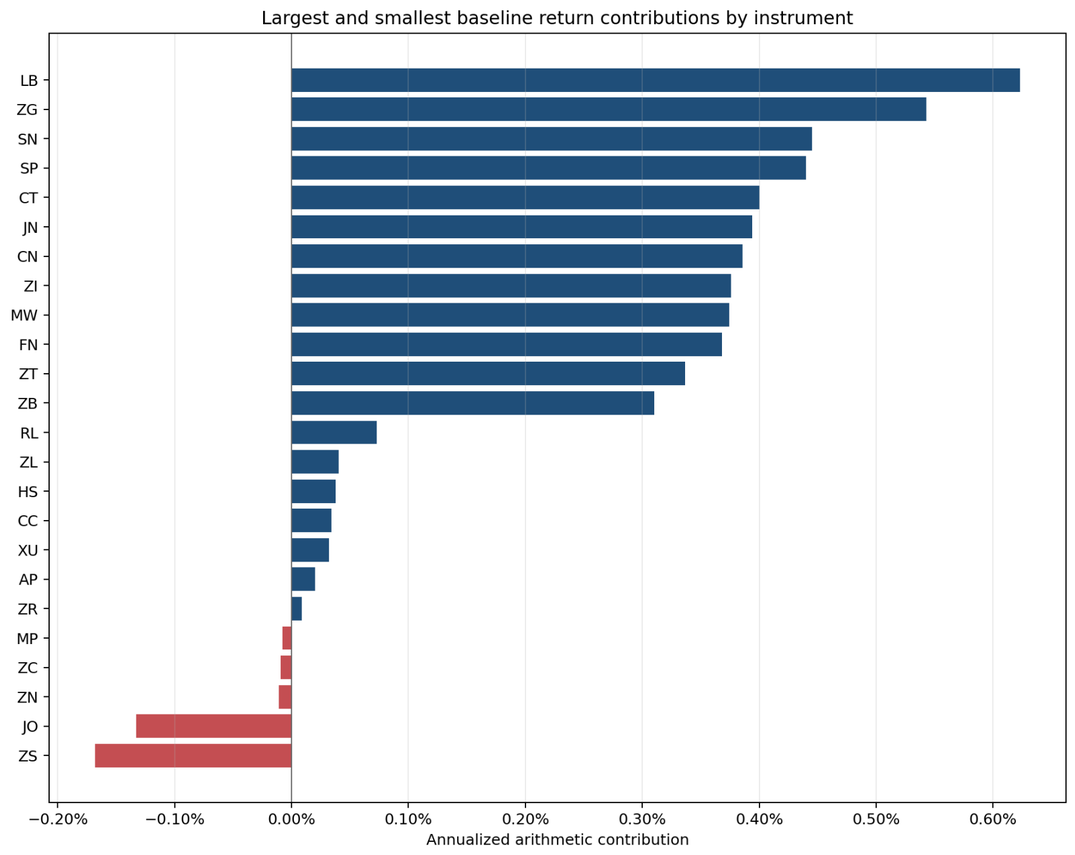
Peak: Feb 2009 Trough: Jul 2009 Recovery: Dec 2013

Peak-to-recovery: 58 months Trough-to-recovery: 53 months

Drawdown = wealth / running peak − 1. Recovery is the first return to that prior peak. Rolling metrics use a moving historical window; they differ from rolling volatility used to form weights.

baseline_performance_output/rolling_performance.csv; drawdown_events.csv: event 57

Return attribution: sources and drawdown drivers



- Commodities: 47.8% of mean return; longs: 81.2%.
- All classes and both directions contribute positively.
- Lumber and gold drive gains; soybeans and orange juice detract most.
- Worst drawdown: AUD, copper, and CAD are major negative contributors.

Monthly contribution $c_{i,t}$ = executed weight × market return. Arithmetic contributions add to portfolio returns; event sums do not decompose compounded MaxDD exactly.

contribution_by_instrument.csv; contribution_by_asset_class.csv; contribution_by_direction.csv; drawdown_event_attribution.csv

Market regimes: definition and attribution

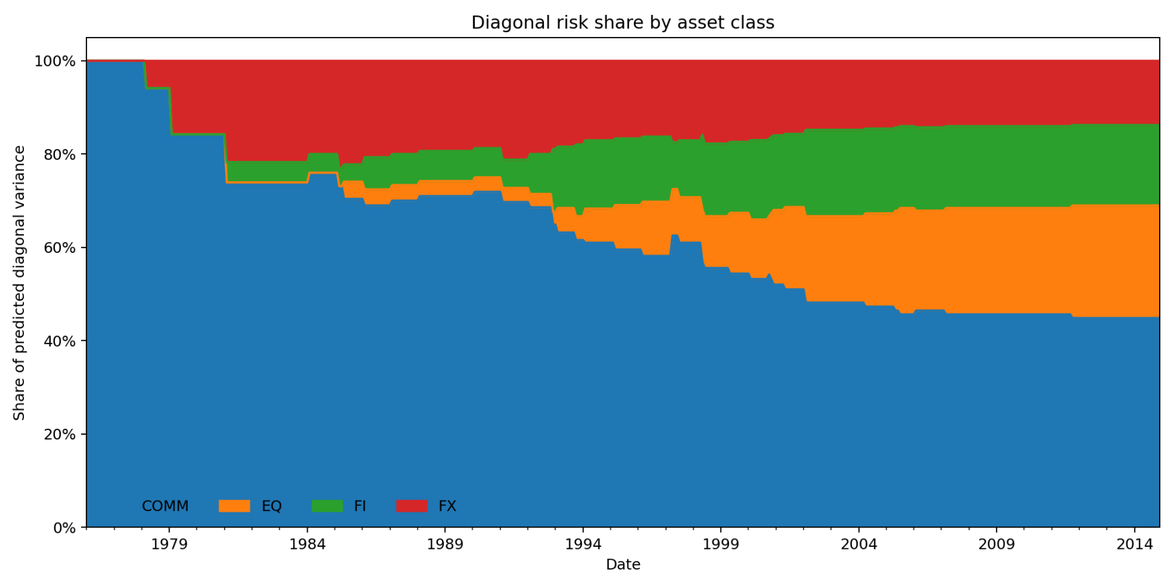
Adjusted baseline	Bear	Bull
Annualized mean	10.26%	11.13%
Annualized volatility	12.60%	8.33%

Classification: compound SP futures returns into proxy wealth. If a peak-to-recovery episode reaches a drawdown of at least 20%, label months after the peak through recovery as Bear. If unrecovered, label through sample end. All remaining observed months are Bull.

Decade mean returns: 1970s 10.37%, 1980s 12.22%, 1990s 9.55%, 2000s 12.29%, 2010–2014 7.57%.

Retrospective futures-based labels, not official cash-index regimes or a real-time trading rule.

Risk concentration under the diagonal model



- Predicted variance shares: commodities 62.7%, FX 15.9%, FI 11.0%, EQ 10.4%.
- Effective risk positions: 39.6 instruments, 2.35 classes.
- Gross exposure sums |weights|; net exposure sums weights.

Diagonal variance = Σ $w_i^2\sigma_i^2$, omitting all cross-market covariance terms. Risk share $q_i = w_i^2\sigma_i^2$ / total; effective positions = $1 / \Sigma q_i^2$.

Validate risk forecasts using fixed weights

Horizon	Predicted vol	Realized vol	Bias	Correlation	Exceedance
12 months	6.82%	9.78%	+2.96pp	0.13	74.8%
3 months	6.79%	8.87%	+2.08pp	0.08	54.6%

Save formation-month prediction. Freeze those weights and measure annualized volatility over the next H months. Error = realized − predicted; bias = mean error. Exceedance = realized > predicted.

No portfolio volatility target exists here. This tests forecast exceedance, not target breaches. Repeat the same test for each enhanced risk model.

Risk validation in high-volatility periods

12m forward-risk sample	Predicted	Realized	Bias	RMSE
High volatility	7.00%	18.58%	+11.58pp	12.23pp
Other periods	6.79%	8.42%	+1.63pp	3.22pp

High-volatility label: future fixed-weight realized vol at or above its 80th-percentile cutoff, separately for each horizon.
This is an ex-post evaluation label, not a forecast available at formation.
In the reported high-volatility 12m subset, realized risk exceeds the prediction by more than 25% in 100% of observations.

forecast_accuracy_high_volatility.csv; validate_diagonal_risk_model.py

SG Trend: external benchmark and active risk

2000–2014	Adjusted	SG Trend
CAGR / Sharpe	10.37% / 0.84	6.90% / 0.53
Annual vol / MaxDD	12.83% / −26.03%	14.60% / −17.66%

Aligned monthly-return comparison	Result
Correlation / raw-return beta	0.616 / 0.54
Annual regression alpha / HAC t	6.52% / 2.48
Tracking error / information ratio	12.12% / 0.245

Tracking error = Std(TSMOM − SG) × √12. Information ratio = annual mean difference / tracking error. Alpha is a regression intercept; beta measures benchmark exposure.

Markets, fees, and risk targets differ. Volatility matching is retrospective; static comparisons remain in Parts 2–3.

sg_trend_comparison.csv; sg_trend_regression.csv; benchmark_methodology.md

Enhancements and next evaluation

Experiment	Gross Sharpe	Matched-vol MaxDD	Monthly turnover
Baseline	0.97	−26.03%	0.60
E1: EWMA, 6m half-life	1.12	−16.63%	1.10
E2: equal class allocation	0.82	−20.81%	0.79
E3: average 1/3/6/12m signs	0.97	−16.46%	1.31
E4: combined	0.87	−14.46%	2.82

- Next: covariance forecasts, portfolio targeting, leverage limits, realistic costs, walk-forward tests.
- Repeat baseline forecast validation for each enhanced risk model.

Matched-vol results use full-sample scaling to baseline volatility, for retrospective comparison. E2 equalizes class allocation, not covariance-aware risk contributions.

strategy_backtest_adj_output/experiment_summary.csv; midterm_report.md